

RESEARCH STATEMENT

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INTRODUCTION

My research interests involve the analysis and implementation of Physics-Informed Neural Networks (PINNs) on nonlinear partial differential equations (PDE), particularly for the simulation of fluid dynamics and other physically motivated systems. While interest in Machine Learning based methods is exploding, rigorous analytic results about the reliability and convergence of these methods for the simulation of PDE remain scarce. Understanding of what makes a differential equation difficult to simulate and when these difficulties arise is essential for the development of robust new methods to assist with PINNs. PINNs are promising but still face unresolved difficulties. This is particularly true for time-dependent problems, which exhibit fundamental difficulties not present in static settings, largely due to the local nature of the PINN training process. In joint work with Roger Temam, we propose a framework for studying PINN convergence via classical PDE well-posedness (existence, uniqueness, and continuity). We additionally apply this analysis on the Cahn-Hilliard and Navier-Stokes Cahn-Hilliard systems.

Additionally I developed Auto-Adaptive sampling (AA-PINN), a training method motivated by this analysis. This method operates on the premise that the choice of sampling distribution in the PINN training process is essential to the accurate simulation of problems with high irregularity. The Auto-Adaptive PINN is a training method which iteratively updates the sampling distribution alongside the network training in proportion to a pre-specified heuristic which may depend on the network itself and its derivatives. We show that this methodology significantly improves training on multiscale problems, in particular the Allen-Cahn equation.

Outside my interest in PINNs, I have broad interest in numerical analysis and physically motivated PDEs. Under the mentorship of Kevin Zumbrun, I have worked on various problems relating to finite difference and finite element methods in the context of shock formation and stability. In particular we are currently working on studying two models. First, a model for Vasculogenesis which shows potential for stable steady state solutions of with Barenblatt-like qualities. We additionally study of the formation of shocks and their spectra in a compressible Navier-Stokes system in a tube.

BACKGROUND ON PINNS

A Physics-Informed Neural Network (as introduced in [1]) is a Neural Network which leverages a partial differential equation in place of labeled training data. Precisely we construct a neural network, which is a composition of many affine and nonlinear functions, and depends on many interior parameters. We construct this network, which we call u_θ , to take inputs x and t , which will be in the domain of our PDE.

$$u_\theta(x, t) := A_n(\sigma(A_{n-1}(\sigma(\dots \sigma(A_1 x + b_1)))) + b_{n-1}) + b_n \quad (1)$$

Here A_n are matrices and b_n are vectors of matching length. We denote these parameters collectively as θ , which are called the hidden layers of the network. The function σ is a nonlinear activation function. This function is chosen depending on the context of the problem but is often either a sigmoid type or rectified linear unit (or ReLU) function.

We then select the parameters θ by optimizing the network to conform to a given differential equation. Take an equation of the form

$$\begin{cases} LHS[u] = RHS(x) \text{ in } \Omega \times (0, T) \\ u(0, x) = u_0(x) \text{ in } \Omega \\ u(x, t) = g(x, t) \text{ on } \partial\Omega \times (0, T), \end{cases} \quad (2)$$

where $LHS[u]$ is a differential operator representing the left hand side of a PDE while $RHS(x)$ represents a forcing function on the right hand side. The second and third equations represent the initial and boundary conditions respectively.

We measure the conformity of the network to the equations by observing the value of the following loss function, which measures the size of the residual of the equations.

$$L(\theta) = \|LHS[u_\theta(x, t)] - RHS(x)\|_{\Omega \times (0, T)}^2 + \|u_\theta(0, x) - u_0(x)\|_{\Omega}^2 + \|u_\theta - g\|_{\partial\Omega \times (0, T)}^2. \quad (3)$$

By minimizing a Monte-Carlo approximation of this loss function with respect to θ , we ‘train’ the network to become an accurate approximation of the solution to the equation (2).

This framework is appealing for its generality, ease of computing derivatives, and lack of spatial discretization. However the method also has many drawbacks which still need to be addressed. Two primary issues are the long training times and the unreliability of the results. Since θ must be taken very large to ensure accuracy, the optimization of the network is an extremely high dimensional optimization problem. This can lead to excessive training times, especially when compared to finite difference or finite element methods for the same problem. Additionally, once the optimization process is complete, there is a reliability issue. Since the final approximation never has residual exactly zero, there is significant room for error. This is especially true in very stiff or otherwise difficult problems. Put simply, a low residual does not guarantee accuracy of similar magnitude. In many cases the accuracy is many order of magnitudes larger than the residual loss.

A final strength to mention here is the potential of this method for improvement. Through methods known as operator learning, it is possible to construct a network which approximates a solution operator instead of a particular solution [2]. This would allow for a network which has information for the approximations of a large class of problems rather than a single one. This would justify the large training times of the network. Additionally, transfer learning is a promising tool to alleviate the training time issue [3]. By starting from a solution to a similar equation, one can hope to greatly reduce the training time of the PINN. Finally, the PINN framework allows very directly for actual data to be incorporated alongside the PDE. This has showed immense promise in the context of inverse problems, but is generally very promising for any situation where one would hope for an approximation which both approximates the solution of a PDE and adheres to empirical data.

PAST PROJECTS

My past work can be split into two main projects as well as several smaller projects. The first primary project involves defining and analyzing the convergence properties of a PINN, and the second constructs an improved training method for particular classes of problems based on the idea of using an adaptive sampling distribution dependent on the networks current state.

Consistency and Convergence

In joint work with Roger Temam, we establish and apply a framework for studying the reliability and effectiveness of a PINN by leveraging commonly used tools in studying well-posedness of PDEs [4]. Precisely, we aim to prove upper bounds on the error given a fixed value of the Loss function. This amounts to a guarantee of accuracy depending on the effectiveness of the training procedure. This stands in contrast to similar error analysis methods which tend to focus on the optimization methods themselves and rely on statistical properties of the network rather than analyze the PDE directly.

We express this in the language of operators. Given the equation (2), the solution operator is given by

$$S(RHS, u_0, g) = u. \quad (4)$$

This solution operator takes inputs the forcing function, initial condition, and boundary condition, and outputs the solution to the PDE. Given that the network approximation u_θ is (and therefore also the Loss (3)) we can define $\delta RHS := LHS[u_\theta] - RHS$, $\delta u_0 := u_\theta(0, \cdot) - u_0$, and $\delta g := u_\theta|_{\partial\Omega} - g$. Then the network u_θ is a solution to the following perturbed equations:

$$\begin{cases} LHS[u_\theta] = RHS(x) + \delta RHS(x) & \text{in } \Omega \times (0, T) \\ u_\theta(0, x) = u_0(x) + \delta u_0(x) & \text{in } \Omega \\ u_\theta(x, t) = g(x, t) + \delta g(x, t) & \text{on } \partial\Omega \times (0, T). \end{cases} \quad (5)$$

Expressed using the notation of (4), this becomes

$$S(RHS + \delta RHS, u_0 + \delta u_0, g + \delta g) = u_\theta \quad (6)$$

Finally, by decomposing u_θ into $u + \delta u$, our goal becomes to show the smallness of δu given the smallness of δRHS , δu_0 , and δg . This is exactly equivalent to the continuity of this solution operator. Improved degrees of Hölder Continuity can then correspond to better convergence of the error of the network to 0 as the loss decreases.

This framework allows us to leverage tools typically used in existence, uniqueness, energy, and continuity methods in order to establish accurate bounds. With this in mind we define consistency and convergence of a PINN. Borrowing from the terminology in more traditional numerical analysis, we define the consistency as the minimal property required for the use a PINN to be sensible for a given problem. That is, that the error approaches zero in the limit as the value of the loss approaches zero.

The convergence of a network is a stricter property, which amounts to the Hölder continuity of the solution operator mentioned below (6). Precisely we define convergence of order n as

$$\|u - u_\theta\|_{\Omega \times (0,T)} \leq C\epsilon^n \quad (7)$$

where we call C the condition number. This property gives considerably more information about the problem. Primarily, this gives a concrete error bound for a PINN simulation given a particular value of the Loss function. This can also be used to diagnose how small the loss value needs to be to achieve a particular level of accuracy. This can also be used to classify the difficulty of a problem. A difficult problem may have a lower order of convergence or a particularly large condition number C .

We use this framework on the particular cases of the Cahn-Hilliard and the Navier-Stokes Cahn-Hilliard equations. We successfully proved the convergence of order 1 of the Cahn-Hilliard system in general and the consistency for the Navier-Stokes Cahn-Hilliard system. It is notable that these properties are proved for general systems with any well-posed initial condition. Given a particular choice of initial condition, a precise condition number could be computed to learn the relative difficulty of the problem. We additionally perform some rudimentary numerical simulations to approximate the order of convergence of the Navier-Stokes Cahn-Hilliard system in one particular test case.

Auto-Adaptive PINNs

In the use of PINNs for the simulation of static equations, oversampling regions which are known to be difficult has proven to be an immensely powerful tool for improving the accuracy of the simulation [5]. For time-dependent problems, it is impossible to reliably specify problematic regions at the beginning of training. If a problematic region develops during training, the spatiotemporal location of the region can also change throughout the training process. These issues make oversampling difficult to implement on time dependent problems, especially problems with low regularity.

In this context we propose an automatically adaptive sampling method which allows for the training process to automatically sample according to quantities known to be highly predictive of error, which we call the Auto-Adaptive PINN (AA-PINN), and show the improved accuracy yielded by this method on a few test cases. The method replaces the Lebesgue norm in (3) with a weighted norm depending on the approximation u_θ . The normalized weight is then used as a probability density for sampling during the training process. This allows for the weighted norm to be computed much more efficiently than if weights were computed directly. It also allows for the use of density functions with very small regions of support, such as interfaces. The sampling is performed in tandem with the training process through the Metropolis-Hastings algorithm, which allows for the sampling of even very complex probability distributions via an iterative method. By performing the iterations of the Metropolis-Hastings method in step with the iterative optimization method used in network training we allow for the sample points to be gradually updated as the network changes without enormous computational expenses.

There are two methods that are worth contrasting to our proposed method. The first is residual or loss adaptive learning [6, 7]. This is a class of method which samples more heavily in regions of high loss. Even outside of PINNs, this method has shown to be very effective for the minimization of the loss. However, it relies on the assumption that an optimal loss is uniform. In the case of highly irregular PDEs, it may still under-resolve narrows, critical sets where low residual still yields large error. The second is the XPINN [8, 9, 10]. This method relies on the decomposition of the domain of a PDE before the simulation begins. The idea here is to split the computational load among the regions, which are then put back together using a stitching condition. This could help in the cases where an AA-PINN

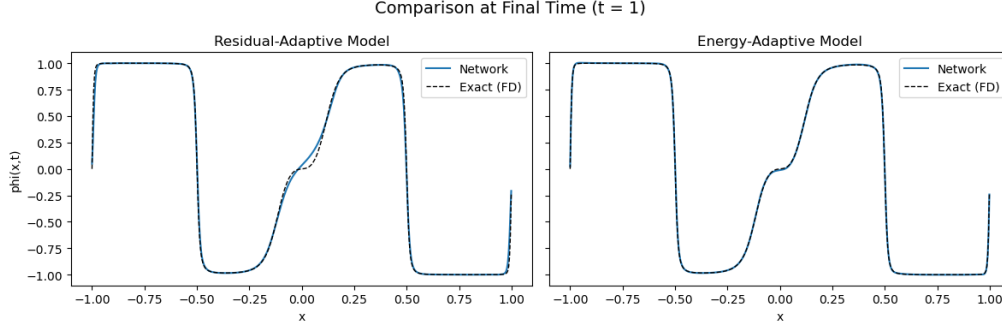


Figure 1: Time slices of a network trained to approximate the solution to a 1D Allen-Cahn Benchmark problem using a Residual-Adaptive method (left) and Energy-Adaptive method (right).

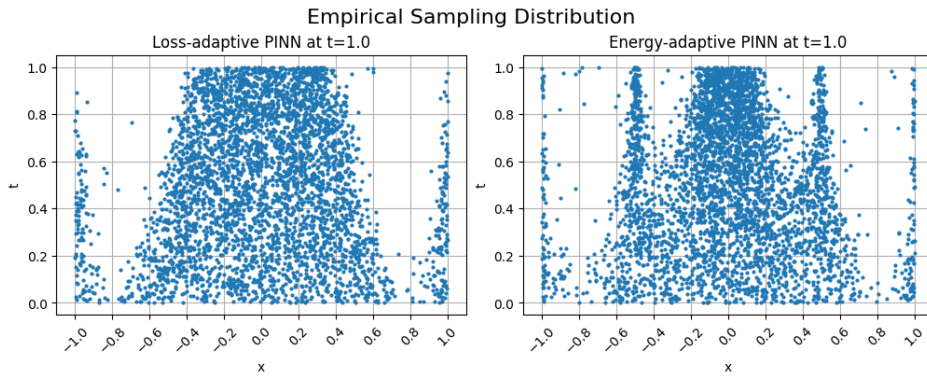


Figure 2: Snapshot of the location of the sample points during the training of a Residual adaptive method (left) and the Energy adaptive method (right) we employ. Notice the energy adaptive sampling points more accurately diagnosing the problematic regions.

is applicable, however it requires a priori knowledge of when and where the problematic regions will occur. The AA-PINN methodology meanwhile only requires a heuristic for identifying the problematic regions and then addresses them as they occur in the training process.

The Allen-Cahn equation was an ideal first test case for this method, as the energy of the system is expensive to compute but is a good indicator of difficult to simulate regions. This equation is additionally characterized by the formation of steep interfaces and the importance of these interfaces to the accuracy of long-time solutions. PINNs have struggled to accurately compute approximations to the solutions of Allen-Cahn in several instances, largely due to the prevalence of these difficulties [11]. We show that by sampling in the described method while using the pointwise energy density as a sampling distribution, accuracy can be greatly increased and we can model the time evolution of these equations. Observe the improvement of the final time slice by the Energy-Adaptive method in Figure 1. We also show the collocation points for each method in Figure 2. Note how the energy-adaptive method more accurately captures the moving formation of the interfaces and identifies the problematic region in the center.

Other Projects

- **Nonconvex Optimization.** [12] We study rigorously from a mathematical point of view Gradient Descent and Stochastic Gradient Descent and discuss the advantage provided by the stochastic variant. We additionally provide the interesting analogue to a continuous time approximation with the Fokker-Planck equations and provide several

numerical examples, including examples in Game Theory.

- **Guts and Game Theory.** [13, 14] Over the course of two summer REU projects, we study the multiple-round game of Guts Poker. In particular we note the distinction between synchronous and asynchronous coalitions in multiplayer games and provide in depth numerical examples and methodologies for the efficient study of these games.

ONGOING WORK

- **AA-PINNs for Higher Order Systems.** An obvious extension of the energy adaptive variant of the Auto-Adaptive PINNs is to the Cahn-Hilliard system. This system shares similar difficulties to Allen-Cahn but with the added complexity of the higher order terms, which can be problematic in themselves.
- **Vasculogenesis.** We study the existence of stable steady state solutions to a model for Vasculogenesis. In particular, we numerically observe and then analytically verify the existence and linear stability of Barenblatt-type solutions.
- **Shock Tube.** We study the formation of shocks of the compressible Navier-Stokes system in a tube. In particular we aim to numerically resolve boundary layer, find stationary solutions, and study the spectra of these solutions.

FUTURE DIRECTIONS

- **Time-Adaptive AA-PINNs.** A major open challenge is the treatment of strongly time-dependent phenomena, particularly ‘switching-time’ problems, which evolve slowly for long periods and then undergo rapid transitions. Extending the Auto-Adaptive framework to time adaptivity could allow the network to concentrate sampling near these transitions, addressing a key failure mode of existing PINNs.
- **Application of the AA-PINN to General Systems.** The convergence framework naturally suggests that one can compute ‘optimal’ sampling distributions for any equation for a broad class of PDE. Implementing these distributions within the Auto-Adaptive framework could create a blueprint for problem-specific neural solvers that maintain accuracy while drastically reducing training cost.
- **Analysis of Loss Function Choice.** Recent work [activation] makes more explicit the intrinsic relationship between the choice of network activation function and the effectiveness of a PINN. We aim to make these notions rigorous and ideally deduce a methodological way to determine when various choices of activation function would be effective.

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